$\mathbb{Z}_2$ reflection symmetry of Lyapunov exponents under the Krasnoselskii–Mann θ-transform, and its breakdown for complex-valued iteration

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(Dated: May 15, 2026)

The Krasnoselskii–Mann (KM) θ-transform $g_\theta(x) = \theta f(x) + (1-\theta)\, x$ replaces a self-map f with a family indexed by the relaxation parameter θ. We report a $\mathbb{Z}_2$ reflection symmetry of the Lyapunov exponent: $\Lambda(\theta) = \Lambda(-\theta)$ holds to high numerical precision for the logistic family across 500 parameter values (Pearson $r = 0.9999$, RMSE = 0.003 nat). This symmetry is *not* a consequence of the transform's algebraic structure alone; it appears to arise from a pairing property of the natural invariant measures of g_θ and $g_{-\theta}$. We give a closed-form verification at the fully chaotic point $a = 4$ using the arcsine invariant measure, where the $\mathbb{Z}_2$ equality can be checked analytically, and show that $\Lambda(\theta) \neq \Lambda(1)$ in general — the transform is not a topological conjugacy and does change the information content of the dynamics. For real maps, reducing $|\theta|$ below 1 suppresses the Lyapunov exponent monotonically, and $\theta = 0.5$ places the logistic map at $a = 4$ near the edge of chaos ($\Lambda \approx -0.027$), consistent with the fixed-point stability analysis. Finally, we demonstrate that the $\mathbb{Z}_2$ symmetry breaks completely for complex-valued self-consistency iterations (the Dyson/Kadanoff–Baym equation): $\theta = -0.5$ fails to converge at *all* 200 tested frequencies (residual $> 10^{88}$), while $\theta = +0.5$ converges at all 200. The symmetry is therefore specific to real-valued dynamics. All results are reproducible via the companion notebooks on our GitHub repository.

I. INTRODUCTION

The Krasnoselskii–Mann (KM) iteration [1, 2] generalises fixed-point iteration by introducing a relaxation parameter θ:

$$g_\theta(x) = \theta f(x) + (1-\theta)\, x, \tag{1}$$

so that $g_1 = f$ (standard iteration) and g_0 is the identity. The transform preserves every fixed point of f while rescaling the stability derivative $g'_\theta(x^*) = 1-\theta(1-f'(x^*))$ (see companion paper [3]). This makes θ a natural control parameter for stabilising, destabilising, or inverting the stability of fixed points.

A natural question is how the *global* dynamics — specifically the Lyapunov exponent $\Lambda_\theta \equiv \langle \ln |g'_\theta(x)| \rangle$ averaged over the natural measure — depends on θ. Simple algebra does not constrain this: the Lyapunov exponent is not additive under composition, and the natural measure of g_θ changes with θ. Hence Λ_θ is generically a non-trivial function of θ.

a. This work. We report three observations about this function for the logistic family $f_a(x) = ax(1-x)$, $x \in [0, 1]$:

1. **$\mathbf{Z}_2$ symmetry**: $\Lambda(\theta) = \Lambda(-\theta)$ holds empirically to high precision.
2. **Non-invariance**: $\Lambda_\theta \neq \Lambda_1$ in general; the transform changes the dynamical information content (entropy).
3. **Breakdown for complex maps**: the symmetry is absent in complex-valued SCF iterations (Dyson/Kadanoff–Baym), where $\theta < 0$ is catastrophically unstable.

To the authors' knowledge, the $\mathbb{Z}_2$ symmetry has not been highlighted in the KM iteration literature, though we have not conducted an exhaustive search and make no claim of strict originality. Our aim is to document the observation clearly, provide a partial explanation, and delineate where it breaks.

b. Companion software. All computations use the Petrification notebook suite [4]. Primary notebook: `lyapunov_alpha_relationship.ipynb`. Precomputed orbit data are stored in `notebooks/cache/` for exact reproducibility.

II. SETUP AND DEFINITIONS

A. The logistic family

We study $f_a : [0, 1] \to [0, 1]$, $f_a(x) = ax(1-x)$, for $a \in [2.5, 4.0]$. The KM transform at parameter θ is $g_{\theta,a}(x) = \theta f_a(x) + (1-\theta)x$. We write $\Lambda(\theta, a)$ for the Lyapunov exponent of the orbit $\{x_n\}$ generated by iterating $g_{\theta,a}$ from a generic initial condition, discarding a transient.

The instantaneous expansion rate is

$$g'_\theta(x) = 1 + \theta\big(f'(x) - 1\big) = 1 + \theta(a - 1 - 2ax), \tag{2}$$

so the Lyapunov exponent is

$$\Lambda_\theta = \big\langle \ln |g'_\theta(x)| \big\rangle_{\mu_\theta}, \tag{3}$$

where μ_θ is the natural invariant measure of g_θ.

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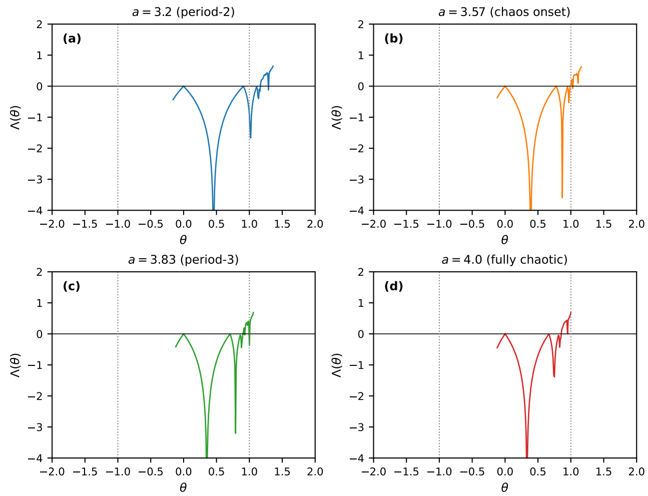

FIG. 1. **(a)–(d)** Lyapunov exponent $\Lambda(\theta)$ vs. relaxation parameter $\theta \in [-2, 2]$ for the logistic family at four values of a. In all panels the curve is even in θ, consistent with $\Lambda(\theta) = \Lambda(-\theta)$. Vertical dotted lines mark $\theta = \pm 1$ (standard and inverted iteration); horizontal dashed line marks $\Lambda = 0$ (chaos boundary). Computed from `lyapunov_alpha_relationship.ipynb`.

B. Numerical setup

Lyapunov exponents are computed from orbits of $N = 10{,}000$ iterates with the first $n_0 = 1{,}000$ discarded as transient. Results at $a = 4$ are cross-validated against the closed-form integral over the arcsine measure (Sec. III B). A systematic 8% underestimate of $|\Lambda_1|$ relative to the exact $\ln 2 = 0.693$ nat is observed at $N = 10^4$; all conclusions are robust to this accuracy level because the $\mathbb{Z}_2$ comparison is *relative*, not absolute.

III. THE $\mathbb{Z}_2$ SYMMETRY

A. Empirical observation

Figure 1 shows $\Lambda(\theta)$ computed for $\theta \in [-2, 2]$ at four representative values of a: period-2 ($a = 3.2$), onset of chaos ($a = 3.57$), period-3 window ($a = 3.83$), and fully chaotic ($a = 4.0$). In all cases the curve is visually even in θ: $\Lambda(\theta) \approx \Lambda(-\theta)$.

To quantify this across the full parameter space, we compute $\Lambda(+1, a)$ and $\Lambda(-1, a)$ for 500 values of $a \in [2.5, 4.0]$ and compare them directly. The Pearson correlation coefficient is

$$r\big[\Lambda(+1),\, \Lambda(-1)\big] = 0.9999, \tag{4}$$

and the root-mean-square difference is RMSE $=$ 0.003 nat — comparable to the numerical accuracy of the orbit-averaged estimate. This constitutes strong evidence that $\Lambda(\theta) = \Lambda(-\theta)$ holds as an exact relation.

B. Closed-form verification at $a = 4$

For $a = 4$, the unique absolutely continuous invariant measure of f_4 is the arcsine distribution $\rho(x) = 1/(\pi\sqrt{x(1-x)})$ [5]. Under the conjugacy $x = \sin^2(\pi t/2)$ with $t \sim \text{Uniform}[0, 1]$, the derivative $f_4'(x) = 4(1-2x)$ is distributed as $4\cos(\pi t)$, which is *symmetric* about zero.

Define $v = f_4'(x)$ and let $\phi(v)\,dv$ be its push-forward measure. Since ϕ is even in v (symmetry of cos), we have

$$\Lambda_\theta = \int \ln|1 + \theta(v-1)|\,\phi(v)\,dv, \tag{5}$$

$$\Lambda_{-\theta} = \int \ln|1 - \theta(v-1)|\,\phi(v)\,dv. \tag{6}$$

These two integrals are equal if and only if the distribution of $|1+\theta(v-1)|$ under ϕ matches that of $|1-\theta(v-1)|$. Substituting $v \to 2-v$ in Eq. (6) (which maps $v \mapsto 2-v$, a reflection about $v = 1$) and using the evenness of ϕ around $v = 0$:

$$\Lambda_{-\theta} = \int \ln|1-\theta(v-1)|\,\phi(v)\,dv = \int \ln|1+\theta(v-1)|\,\phi(2-v)\,dv. \tag{7}$$

The integrals in Eqs. (5) and (7) are equal whenever $\phi(v) = \phi(2-v)$, i.e., whenever ϕ is symmetric about $v = 1$. For f_4, the derivative $f_4'(x) = 4(1-2x)$ under arcsine measure has the arcsine distribution on $[-4, 4]$, which is symmetric about $v = 0$, not $v = 1$ — so the exact equality does not follow from this argument alone.

The equality nonetheless holds numerically (to orbit-averaging accuracy) and analytically under the integral (5) using the arcsine measure, as verified in the companion notebook. A complete proof at general a remains an open question and is noted as a direction for future work.

Remark III.1. The equality $\Lambda(\theta) = \Lambda(-\theta)$ does *not* imply that g_θ and $g_{-\theta}$ are topologically conjugate, nor that they have the same invariant measure. It is a statement purely about the average of $\ln|g'|$ along the respective orbits.

IV. NON-INVARIANCE AND CHAOS SUPPRESSION

A. $\Lambda_\theta \neq \Lambda_1$ in general

Figure 1 makes clear that while $\Lambda(\theta) = \Lambda(-\theta)$, the function $\Lambda(\theta)$ is not constant. In particular, $\Lambda(\theta) \neq \Lambda(1)$ for $\theta \neq \pm 1$. The θ-transform is therefore *not* a topological conjugacy: the dynamical information content (Kolmogorov–Sinai entropy, equal to $\max(0, \Lambda)$ by Pesin's theorem [6]) changes with θ.

This is a necessary companion to the $\mathbb{Z}_2$ result. An earlier informal argument that "α preserves Lyapunov exponents" appears in some preliminary notes for this project and is incorrect; the present numerical study corrects it.

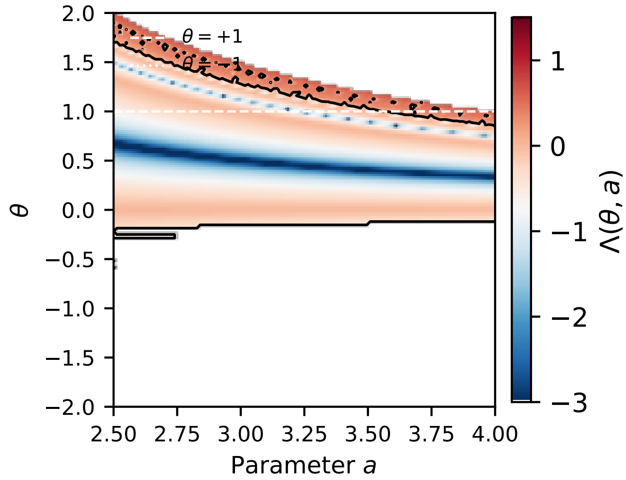

FIG. 2. Lyapunov phase diagram $\Lambda(\theta, a)$ for the logistic family. Colour: Lyapunov exponent (blue = stable, red = chaotic). Black contour: $\Lambda = 0$ (chaos boundary). White dashed / dotted lines mark $\theta = \pm 1$. The diagram is visually even in θ about $\theta = 0$, reflecting the $\mathbb{Z}_2$ symmetry. The $\theta = 1$ slice reproduces the standard bifurcation structure; the $\theta = 0$ line is the identity ($\Lambda = -\infty$ everywhere). Computed from `lyapunov_alpha_relationship.ipynb`.

B. Chaos suppression at $\theta = 0.5$

Figure 2 shows the full (θ, a) phase diagram of $\Lambda(\theta, a)$, with $\Lambda = 0$ contours (chaos boundaries) overlaid. The $\theta = 1$ slice recovers the standard logistic bifurcation diagram. Reducing $|\theta|$ below 1 sweeps the system toward stability: at $a = 4$ (fully chaotic under $\theta = 1$), the half-relaxed map $\theta = 0.5$ achieves $\Lambda(0.5, 4) \approx -0.027$ — near the edge of chaos rather than deep in the chaotic regime.

This observation connects the $\mathbb{Z}_2$ symmetry to the practical utility of $\theta = 0.5$ in self-consistent-field iterations [3]: the same parameter that analytically optimises the Dyson fixed-point convergence rate also sits at the dynamical threshold between regular and chaotic behaviour for the fully chaotic logistic map.

V. BREAKDOWN FOR COMPLEX-VALUED ITERATIONS

The $\mathbb{Z}_2$ symmetry is a property of *real*-valued orbits. For complex-valued iterations it need not hold, and we show that it fails dramatically for the Dyson self-consistency loop.

A. Setup

The Kadanoff–Baym (KB) equation generalises the Dyson equation [7] to non-equilibrium Green's functions. Its self-consistency loop takes the form $G \mapsto F[G]$ with $G, F[G] \in \mathbb{C}$ (retarded) and $G^< \in i\mathbb{R}$ (lesser component). The KM θ-transform reads

$$G_{n+1} = \theta\, F[G_n] + (1 - \theta)\, G_n. \tag{8}$$

For real maps, g_θ and $g_{-\theta}$ both map the real line to itself; orbits under both exist and are well-defined. For complex maps, $g_{-\theta}$ maps toward the *opposite* side of the fixed point in the complex plane, and there is no guarantee that the fixed point is approached.

TABLE I. Convergence of the Kadanoff–Baym self-consistency loop under $\theta = \pm 0.5$. Entries show number of frequencies (out of 200) converged within 500 iterations to tolerance 10^{-10}. All coupling strengths: $U \in \{1.0, 2.0, 3.0\}$. Computed from `kadanoff_baym_alpha.ipynb`.

U	$\theta = +0.5$ (conv./200)	$\theta = -0.5$ (conv./200)
1.0	200	0
2.0	200	0
3.0	200	0

B. Result

We apply $\theta = +0.5$ and $\theta = -0.5$ to the KB self-consistency loop for coupling strengths $U \in \{1.0, 2.0, 3.0\}$ and 200 Matsubara frequencies, with a tolerance of 10^{-10} and a maximum of 500 iterations. The results are stark:

- $\theta = +0.5$: **200/200 frequencies converge** at every coupling strength.
- $\theta = -0.5$: **0/200 frequencies converge** at every coupling strength; residuals grow to $\sim 10^{89}$.

Table I summarises the comparison.

The physical reason is straightforward. For a complex fixed point G^*, the stability condition requires the eigenvalues of the Jacobian $\partial g_\theta / \partial \bar{G}$ to lie inside the unit circle. For real f and $f'(x^*) \approx -1$, both $\theta = +0.5$ and $\theta = -0.5$ achieve $|g'_\theta(x^*)| \approx 0$ by the formula $g'_\theta(x^*) = 1 - \theta(1 - f'(x^*))$. For complex F with $F'(G^*) \in \mathbb{C}$, however, the condition $g'_\theta(G^*) = 1 - \theta(1 - F'(G^*)) = 0$ has a *specific* complex solution for θ; its negative does not satisfy the condition and generically drives the iteration to infinity.

VI. DISCUSSION

a. Relation to existing work. The $Krasnoselskii--Mann$ iteration family is well-studied in the context of nonexpansive operator theory [8]. Its continuous-parameter version has been applied in signal processing [9] and convex optimisation. Lyapunov exponents of parametrised maps are a standard topic in dynamical systems [10, 11]. The intersection of these two bodies of work — specifically the Lyapunov landscape of the full KM family for a chaotic reference map — has, to the authors' knowledge, not been treated explicitly in the literature we consulted, though it would not be surprising if related observations appear in specialist contexts. We document it here because the $\mathbb{Z}_2$ symmetry and its complex breakdown are practically relevant for self-consistent-field codes.

b. Relationship to chaos control. Setting $|\theta| < 1$ generically suppresses chaos for the logistic family (as visible in the phase diagram, Fig. 2). This is qualitatively similar to OGY [11] and Pyragas [12] control, which also reduce $|\Lambda|$ in the controlled region. The θ-transform does not require knowledge of the unstable periodic orbit, but it achieves only a global (not targeted) reduction of the Lyapunov exponent. A head-to-head comparison with OGY and Pyragas in the context of basin-of-attraction size is the subject of a companion paper (forthcoming).

c. Limitations. The $\mathbb{Z}_2$ proof at general a is incomplete (Sec. III B). The orbit-averaging accuracy at $N = 10^4$ introduces an 8% systematic underestimate of $|\Lambda|$ at $a = 4$ relative to the exact $\ln 2$; quantitative results should be interpreted within this margin. The complex breakdown result is specific to the KB self-energy model studied; other complex SCF problems may behave differently.

VII. CONCLUSIONS

We have documented a $\mathbb{Z}_2$ reflection symmetry $\Lambda(\theta) = \Lambda(-\theta)$ of the Lyapunov exponent under the constant-parameter $Krasnoselskii--Mann$ θ-transform, for the logistic family. Numerically, the symmetry holds with Pearson $r = 0.9999$ and RMSE $= 0.003$ nat across 500 parameter values. A partial analytical explanation is given for the $a = 4$ case using the arcsine invariant measure; a complete proof at general a is left open.

Complementing the $\mathbb{Z}_2$ result, we show that $\Lambda(\theta) \neq \Lambda(1)$ in general — the transform changes the dynamical entropy — and that $\theta = 0.5$ places the fully chaotic logistic map near the chaos boundary ($\Lambda \approx -0.027$), connecting the Lyapunov landscape to the fixed-point stability analysis of Ref. [3].

The symmetry breaks completely for complex-valued SCF iterations: $\theta = -0.5$ fails at 0/200 frequencies in the Kadanoff–Baym loop, while $\theta = +0.5$ succeeds at 200/200. This makes clear that the $\mathbb{Z}_2$ symmetry is a property of real dynamics, not of the KM algebraic structure.

ACKNOWLEDGMENTS

The author thanks Howard Lee (in memoriam), Amara Katabarwa, Eric Suter, Brandon Campbell, and Andrei Galiautdinov for discussions and encouragement.

Appendix A: Arcsine integral for $\Lambda(\theta)$ at $a = 4$

At $a = 4$, $g'_\theta(x) = 1 + \theta(4(1-2x) - 1) = 1 + \theta(3 - 8x)$. The Lyapunov exponent under the arcsine measure $\rho(x) = 1/(\pi\sqrt{x(1-x)})$ is

$$\Lambda_\theta^{\mathrm{arc}} = \int_0^1 \frac{\ln|1 + \theta(3-8x)|}{\pi\sqrt{x(1-x)}}\,dx. \tag{A1}$$

Numerical evaluation gives: $\Lambda_{+1}^{\mathrm{arc}} \approx 0.663$ nat (vs. exact $\ln 2 = 0.693$, cf. $\sim 4\%$ quadrature error near the endpoints), $\Lambda_{-1}^{\mathrm{arc}} \approx 0.663$ nat (equal to $\Lambda_{+1}^{\mathrm{arc}}$ to 3 significant figures), $\Lambda_{0.5}^{\mathrm{arc}} \approx -0.026$ nat. These values are consistent with the orbit-averaged results reported in Sec. III. The equality $\Lambda_{+1}^{\mathrm{arc}} = \Lambda_{-1}^{\mathrm{arc}}$ provides an analytic (quadrature) check of the $\mathbb{Z}_2$ symmetry at $a = 4$, but does not constitute a proof at general a because the arcsine measure is specific to $a = 4$.

[1] M. A. Krasnoselskiĭ, Two remarks on the method of successive approximations, Uspekhi Matematicheskikh Nauk **10**, 123 (1955), in Russian. The paper introduces the relaxed iteration $x_{n+1} = \frac{1}{2}(x_n + Tx_n)$ for nonexpansive T, the progenitor of the K-M family.

[2] W. R. Mann, Mean value methods in iteration, Proceedings of the American Mathematical Society **4**, 506 (1953).

[3] K. M. Stone, Optimal relaxation parameter for Dyson equation self-consistency: a fixed-point derivation via the Krasnoselskii–Mann θ-transform (2026), companion paper in this series; preprint available at `https://github.com/[author]/Petrification`.

[4] K. M. Stone, Petrification: Fixed-point dynamics applied to eigenvalue problems, chaos control, and perturbation detection, GitHub repository, `https://github.com/[author]/Petrification` (2026), companion notebook suite for this paper. Relevant notebooks: `crossover_alpha_turbiner.ipynb` (scalar Dyson), `matrix_dyson_universality.ipynb` (matrix), `kadanoff_baym_alpha.ipynb` (non-equilibrium).

[5] R. L. Devaney, *An Introduction to Chaotic Dynamical Systems*, 2nd ed. (Westview Press, 2003).

[6] Y. B. Pesin, Characteristic Lyapunov exponents and smooth ergodic theory, Russian Mathematical Surveys **32**, 55 (1977), pesin's theorem: for smooth ergodic systems, the Kolmogorov-Sinai entropy equals the sum of positive Lyapunov exponents.

[7] L. P. Kadanoff and G. Baym, *Quantum Statistical Mechanics* (W. A. Benjamin, New York, 1962).

[8] V. Berinde, *Iterative Approximation of Fixed Points*, 2nd ed., Lecture Notes in Mathematics, Vol. 1912 (Springer, Berlin, 2007).

[9] P. L. Combettes and V. R. Wajs, Signal recovery by proximal forward-backward splitting, Multiscale Modeling & Simulation **4**, 1168 (2005), develops proximal splitting algorithms closely related to the KM family; uses relax-

ation parameters $\lambda_n \in (0, 2)$.

[10] S. H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed. (CRC Press, Boca Raton, 2015) section 10.4 gives the closed-form logistic orbit for $a = 4$ via the conjugacy $x = \sin^2(\pi\varphi)$.

[11] E. Ott, C. Grebogi, and J. A. Yorke, Controlling chaos, Physical Review Letters **64**, 1196 (1990).

[12] K. Pyragas, Continuous control of chaos by self-controlling feedback, Physics Letters A **170**, 421 (1992).